Year 9 · Chapter 12 · Probability: answers (teacher copy)

12.1 Probability (booklet p. 3)

WE DO examples (teacher)

1. Ex 1a: Total = 10; P(red) = 2/10 = 1/5; P(not red) = 1 − 1/5 = 4/5
2. Ex 1b: Total = 12; P(blue) = 6/12 = 1/2; P(not blue) = 1 − 1/2 = 1/2
3. Ex 1c: Total = 7; P(A) = 3/7; P(not A) = 1 − 3/7 = 4/7
4. Ex 1d: Total = 10; 7, 8, 9, 10: P(more than 6) = 4/10 = 2/5; P(not more than 6) = 1 − 2/5 = 3/5
5. Ex 2a: P(yellow) = 100% − 40% = 60% = 60/100 = 3/5
6. Ex 2b: 40% × 40 = 2/5 × 40 = 16 red marbles
7. Ex 2c: 26 red cards (hearts and diamonds): P(red card) = 26/52 = 1/2
8. Ex 2d: 4 Kings + 4 Queens = 8: P(King or Queen) = 8/52 = 2/13

Easy

1a Total = 8; P(red) = 3/8; P(not red) = 1 − 3/8 = 5/8

1b Total = 10; P(green) = 4/10 = 2/5; P(not green) = 1 − 2/5 = 3/5

2a Total = 12; P(blue) = 4/12 = 1/3; P(not blue) = 1 − 1/3 = 2/3

2b Total = 15; P(white) = 7/15; P(not white) = 1 − 7/15 = 8/15

3a Total = 6; P(E) = 2/6 = 1/3; P(not E) = 1 − 1/3 = 2/3

3b Total = 8; P(L) = 3/8; P(not L) = 1 − 3/8 = 5/8

4a Total = 12; P(green) = 4/12 = 1/3; P(not green) = 1 − 1/3 = 2/3

4b Total = 16; P(red) = 2/16 = 1/8; P(not red) = 1 − 1/8 = 7/8

Medium

5 P(heart) = 13/52 = 1/4; P(not heart) = 1 − 1/4 = 3/4

6 18/24 = 3/4 (Year 8)

7 P(yellow) = 1 − 3/8 = 5/8

8 1/4 × 24 = 6 red

9 m = (8 − 2) ÷ (3 − 1) = 3 (11.3)

10 Total = 11; P(M) = 2/11; P(not M) = 1 − 2/11 = 9/11

Worded problems

1. Total = 12 + 18 = 30; P(girl) = 18/30 = 3/5; P(not girl) = 1 − 3/5 = 2/5
2. P(green) = 1 − 6/20 − 9/20 = 5/20 = 1/4; 1/4 × 40 = 10 green
3. P(Mia wins) = 8/500 = 2/125; P(not Mia wins) = 1 − 2/125 = 123/125

12.2 Experimental probability (booklet p. 6)

WE DO examples (teacher)

1. Ex 1a: Total = 50; observed P(head) = 28/50 = 14/25; theoretical P(head) = 1/2
2. Ex 1b: observed P(3) = 12/40 = 3/10; theoretical P(3) = 1/4
3. Ex 1c: Total = 30; observed P(red) = 12/30 = 2/5; theoretical P(red) = 1/3
4. Ex 1d: 1/4 × 200 = 50 times
5. Ex 2a: observed P(red) = 72/200 = 0.36
6. Ex 2b: 29 + 35 = 64, so observed P(blue or green) = 64/200 = 0.32
7. Ex 2c: 6 equal parts: theoretical P(red) = 2/6 = 1/3 ≈ 0.33; theoretical P(not purple) = 5/6 ≈ 0.83
8. Ex 2d: 1/3 × 500 ≈ 166.7, about 167 reds

Easy

1a Total = 40; observed P(red) = 13/40; theoretical P(red) = 1/4

1b Total = 60; observed P(blue) = 24/60 = 2/5; theoretical P(blue) = 1/3

2a Total = 60; observed P(4) = 12/60 = 1/5; theoretical P(4) = 1/6

2b Total = 120; 22 + 21 + 24 = 67, so observed P(even) = 67/120; theoretical P(even) = 3/6 = 1/2

3a Total = 100; observed P(red) = 46/100 = 0.46; theoretical P(red) = 2/4 = 1/2 = 0.5

3b Total = 50; observed P(blue) = 22/50 = 0.44; theoretical P(blue) = 2/5 = 0.4

4a Total = 20; observed P(head) = 12/20 = 3/5; theoretical P(head) = 1/2

4b Total = 15; observed P(blue) = 7/15; theoretical P(blue) = 1/3

Medium

5 Total = 50; observed P(2) = 14/50 = 7/25

6 P(blue) = 2/10 = 1/5; P(not blue) = 1 − 1/5 = 4/5 (12.1)

7 0.35 × 400 = 140 reds

8 1/6 × 600 = 100 sixes expected; 180 is far more than 100, so the die may not be fair

9 7 ÷ 20 = 0.35 = 35% (Year 8)

10 Total = 40; observed P(red) = 18/40 = 0.45 = 45%

Worded problems

1. observed P(makes it) = 36/48 = 3/4; 3/4 × 200 = 150 throws
2. observed P(6) = 71/300 ≈ 0.24; theoretical P(6) = 1/6 ≈ 0.17; expected 1/6 × 300 = 50 sixes, but 71 came up, so it may not be fair
3. observed P(red) = 18/120 = 3/20; 3/20 × 400 = 60 red cars

12.3 Venn diagrams (booklet p. 9)

WE DO examples (teacher)

1. Ex 1a: Total = 20; P(B) = (8 + 4)/20 = 12/20 = 3/5
2. Ex 1b: P(B and S) = 4/20 = 1/5
3. Ex 1c: P(B or S) = (8 + 4 + 6)/20 = 18/20 = 9/10
4. Ex 1d: P(neither) = 2/20 = 1/10
5. Ex 2a: Both = 21 + 18 + 4 − 30 = 13
6. Ex 2b: M only = 21 − 13 = 8, both 13, T only = 18 − 13 = 5, neither 4
7. Ex 2c: P(M only) = 8/30 = 4/15
8. Ex 2d: P(only one) = (8 + 5)/30 = 13/30

Easy

1a Total = 25; P(S) = (7 + 5)/25 = 12/25; P(S and M) = 5/25 = 1/5; P(S or M) = (7 + 5 + 9)/25 = 21/25; P(neither) = 4/25

1b Total = 20; P(C) = (6 + 4)/20 = 10/20 = 1/2; P(C and D) = 4/20 = 1/5; P(C or D) = (6 + 4 + 8)/20 = 18/20 = 9/10; P(neither) = 2/20 = 1/10

2a Total = 30; P(T) = (9 + 6)/30 = 15/30 = 1/2; P(T and C) = 6/30 = 1/5; P(T or C) = (9 + 6 + 12)/30 = 27/30 = 9/10; P(neither) = 3/30 = 1/10

2b Total = 24; P(B) = (11 + 3)/24 = 14/24 = 7/12; P(B and R) = 3/24 = 1/8; P(B or R) = (11 + 3 + 7)/24 = 21/24 = 7/8; P(neither) = 3/24 = 1/8

3a Total = 40; P(N) = (14 + 6)/40 = 20/40 = 1/2; P(N and S) = 6/40 = 3/20; P(N or S) = (14 + 6 + 10)/40 = 30/40 = 3/4; P(neither) = 10/40 = 1/4

3b Total = 50; P(L) = (5 + 27)/50 = 32/50 = 16/25; P(L and P) = 27/50; P(L or P) = (5 + 27 + 12)/50 = 44/50 = 22/25; P(neither) = 6/50 = 3/25

4a Both = 20 + 15 + 5 − 32 = 8; S only 12, M only 7; P(S) = (12 + 8)/32 = 20/32 = 5/8; P(S and M) = 8/32 = 1/4; P(S or M) = (12 + 8 + 7)/32 = 27/32; P(neither) = 5/32

4b Both = 25 + 18 + 6 − 40 = 9; C only 16, D only 9; P(C) = (16 + 9)/40 = 25/40 = 5/8; P(C and D) = 9/40; P(C or D) = (16 + 9 + 9)/40 = 34/40 = 17/20; P(neither) = 6/40 = 3/20

Medium

5 Total = 20; P(A only) = 6/20 = 3/10; P(not B) = (6 + 4)/20 = 10/20 = 1/2

6 Total = 40; observed P(T) = 26/40 = 13/20 (12.2)

7 Total = 20; P(only one) = (9 + 4)/20 = 13/20

8 Both = 15 + 12 + 5 − 24 = 8; P(both) = 8/24 = 1/3

9 P(not club) = 1 − 13/52 = 1/4 = 3/4 (12.1)

10 Total = 40; P(neither) = 5/40 = 1/8; P(T or W) = (12 + 8 + 15)/40 = 35/40 = 7/8

Worded problems

1. N only 15 − 6 = 9, S only 10 − 6 = 4, both 6, neither 28 − 19 = 9: P(neither) = 9/28
2. Both = 32 + 20 + 8 − 50 = 10; P(bread or milk) = (50 − 8)/50 = 42/50 = 21/25
3. E has 10, T has 6, both: 6, 12, 18; P(E and T) = 3/20; P(E or T) = (10 + 6 − 3)/20 = 13/20

12.4 Tree diagrams (booklet p. 12)

WE DO examples (teacher)

1. Ex 1a: 4 outcomes: HH, HT, TH, TT
2. Ex 1b: 4 outcomes: HH, HT, TH, TT; P(two tails) = 1/4 (TT)
3. Ex 1c: 6 outcomes: H1, H2, H3, T1, T2, T3
4. Ex 1d: 6 outcomes: H1, H2, H3, T1, T2, T3; P(tail and odd) = 2/6 = 1/3 (T1, T3)
5. Ex 2a: 4 × 3 = 12: AB, AC, AD, BA, BC, BD, CA, CB, CD, DA, DB, DC
6. Ex 2b: AB, BA: P(A and B) = 2/12 = 1/6
7. Ex 2c: all but CD, DC: P(A or B) = 10/12 = 5/6
8. Ex 2d: P(no D) = 6/12 = 1/2

Easy

1a 4 outcomes: HH, HT, TH, TT; P(two heads) = 1/4 (HH); P(at least one head) = 3/4 (HH, HT, TH)

1b 6 outcomes: RH, RT, BH, BT, YH, YT; P(red and head) = 1/6 (RH); P(a head) = 3/6 = 1/2 (RH, BH, YH)

2a 6 outcomes: MA, MB, MP, JA, JB, JP; P(juice and pear) = 1/6 (JP); P(banana) = 2/6 = 1/3 (MB, JB)

2b 9 outcomes: 1,1, 1,2, 1,3, 2,1, 2,2, 2,3, 3,1, 3,2, 3,3; P(a double) = 3/9 = 1/3 (1,1, 2,2, 3,3); P(a total of 5) = 2/9 (2,3, 3,2)

3a 6 outcomes: AB, AC, BA, BC, CA, CB; P(A then B) = 1/6 (AB); P(no C) = 2/6 = 1/3 (AB, BA)

3b 6 outcomes: 12, 13, 21, 23, 31, 32; P(an even number) = 2/6 = 1/3 (12, 32); P(more than 20) = 4/6 = 2/3 (21, 23, 31, 32)

4a 8 outcomes: H1, H2, H3, H4, T1, T2, T3, T4; P(head and 4) = 1/8 (H4); P(an even number) = 4/8 = 1/2 (H2, H4, T2, T4)

4b 6 outcomes: PW, PJ, PM, SW, SJ, SM; P(salad and milk) = 1/6 (SM); P(not water) = 4/6 = 2/3 (PJ, PM, SJ, SM)

Medium

5 4 outcomes: HH, HT, TH, TT; P(HT or TH) = 2/4 = 1/2

6 Total = 20; P(A and B) = 3/20 (12.3)

7 6 × 2 = 12 outcomes

8 4 × 4 = 16 outcomes; 11, 22, 33, 44: P(double) = 4/16 = 1/4

9 0.3 × 250 = 75 reds (12.2)

10 3 × 2 = 6 outcomes; XY, XZ: P(X first) = 2/6 = 1/3

Worded problems

1. 4 outcomes: BB, BG, GB, GG; P(one of each) = 2/4 = 1/2 (BG, GB)
2. 6 outcomes: RK, RW, BK, BW, GK, GW; P(blue and white) = 1/6 (BW); P(no red) = 4/6 = 2/3 (BK, BW, GK, GW)
3. 6 outcomes: AB, AC, BA, BC, CA, CB; P(Cara chosen) = 4/6 = 2/3 (AC, BC, CA, CB)

12.5 Tables of outcomes (booklet p. 15)

WE DO examples (teacher)

1. Ex 1a: Sums: 2, 3, 4, 3, 4, 5, 4, 5, 6
2. Ex 1b: Total = 3 × 3 = 9; P(sum of 4) = 3/9 = 1/3
3. Ex 1c: Total = 3 × 3 = 9; P(odd sum) = 4/9
4. Ex 1d: Total = 2 × 6 = 12; P(head and 3) = 1/12
5. Ex 2a: 36 cells: 0 1 2 3 4 5 / 1 0 1 2 3 4 / 2 1 0 1 2 3 / 3 2 1 0 1 2 / 4 3 2 1 0 1 / 5 4 3 2 1 0
6. Ex 2b: P(0) = 6/36 = 1/6
7. Ex 2c: P(at least 2) = 20/36 = 5/9
8. Ex 2d: P(4 or 5) = 6/36 = 1/6

Easy

1a Total = 4 × 3 = 12; P(sum of 5) = 3/12 = 1/4; P(even sum) = 6/12 = 1/2; P(sum more than 5) = 3/12 = 1/4

1b Total = 2 × 6 = 12; P(head and even) = 3/12 = 1/4; P(a 6) = 2/12 = 1/6; P(tail or 1) = 7/12

2a Total = 6 × 6 = 36; P(sum of 8) = 5/36; P(sum of 10 or more) = 6/36 = 1/6; P(even sum) = 18/36 = 1/2

2b Total = 6 × 6 = 36; P(product of 12) = 4/36 = 1/9; P(odd product) = 9/36 = 1/4; P(product more than 20) = 6/36 = 1/6

3a Total = 2 × 3 = 6; P(same colour) = 2/6 = 1/3; P(at least one blue) = 4/6 = 2/3; P(no red) = 2/6 = 1/3

3b Total = 4 × 4 = 16; P(product of 4) = 3/16; P(odd product) = 4/16 = 1/4; P(product more than 6) = 6/16 = 3/8

4a Total = 3 × 3 = 9; P(sum of 4) = 3/9 = 1/3; P(sum more than 4) = 3/9 = 1/3; P(odd sum) = 4/9

4b Total = 3 × 4 = 12; P(difference of 0) = 3/12 = 1/4; P(difference of at least 2) = 4/12 = 1/3; P(odd difference) = 6/12 = 1/2

Medium

5 Total = 2 × 2 = 4; P(HT or TH) = 2/4 = 1/2

6 3 × 4 = 12 outcomes (12.4)

7 Total = 6 × 6 = 36; P(sum of 9) = 4/36 = 1/9

8 Total = 6 × 6 = 36; P(both even) = 9/36 = 1/4

9 Total = 20; P(A or B) = (4 + 2 + 5)/20 = 11/20 (12.3)

10 Total = 3 × 3 = 9; P(even product) = 5/9

Worded problems

1. Total = 6 × 6 = 36; P(7 or 11) = 8/36 = 2/9
2. Total = 4 × 5 = 20; P(more than 10) = 5/20 = 1/4
3. Total = 6 × 6 = 36; P(6) = 11/36; P(1) = 1/36